

Liability, Verification, and the Market for Expertise in the Presence of AI

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Abstract

Artificial intelligence makes competent execution cheap, so for tasks a principal can verify himself he increasingly performs them directly rather than hiring an intermediary. For tasks he cannot verify, such as a diagnosis or an audit, hiring remains necessary. But what made hiring valuable, the intermediary's credibility, is exactly what a commoditized AI erodes: once anyone can produce a fluent-looking deliverable, fluency stops being informative. A signal, such as a liability bond, can restore separation between genuine expertise and imitation, but only if the bond is enforceable, and enforceability runs into the same verification problem that created the need for the signal. We build a model that nests a make-or-buy decision inside a liability-bond signaling game and characterize when this tension resolves in the market's favor and when it does not. The central result is that whether the market for genuine expertise can fail at all does not depend on how good the best available expert is. It depends on the quality of the worst type in the market, the principal's own fallback, and how much capital a genuine expert can post. Better experts do not prevent the failure region; they only change its size.

Keywords: liability, signaling, credence goods, verification, artificial intelligence, delegation, judgment-proof problem

JEL codes: D82, D86, K13, L84, O33

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1 Introduction

Should a firm hire a consultant to run an analysis, or run it itself with an AI model? The answer used to turn mostly on capability: could the firm produce a comparable result at all? It increasingly turns on something else: can the firm *tell*, once the work is done, whether it is any good? For a webpage, an image or video, a small software feature, or a first-pass slide deck, the answer is close to yes. The principal can click through it, run it, or compare it against a brief. For a tax opinion on a novel transaction, an audit of a firm’s financial statements, or a medical diagnosis, the answer is closer to no: judging the output requires much the same expertise that produced it. The first class of tasks is drifting toward direct AI use. The second class is where hiring survives, and where a genuine problem is opening up.

The problem is not new in kind. It is Akerlof’s market for lemons, restaged: if a principal cannot verify quality directly, he must rely on some correlate of quality he *can* observe, and the market unravels toward the correlate’s worst case unless a credible signal separates types. What is new is that AI has begun dissolving the signals that markets for expertise have long relied on. A polished deliverable, a fluent report, even demonstrated familiarity with the relevant literature used to correlate with genuine expertise because producing them was costly for anyone who lacked it. A sufficiently capable AI makes them cheap for anyone. These signals are eroding at the same moment verification is becoming harder, because AI-assisted outputs are increasingly sophisticated enough that only genuine expertise can check them. That is precisely the credence-good condition that makes unraveling possible in the first place.

Liability is a classical response to this problem: if quality cannot be verified before the fact, make the seller bear the cost of it being wrong after the fact. A performance bond, forfeited on a provable failure, lets a genuine expert post a credible signal even when the principal cannot inspect the work directly. Courts, not the principal, do the verifying, *ex post*. But this response inherits the same disease it is meant to cure. A court’s ability to determine that a delivered audit or diagnosis was wrong is itself a verification problem, and often a harder one than the principal’s, since the finding must also survive an adversarial process. If it is hard for the principal to tell good work from bad, it will typically be hard for a court to establish it too. In the model below, this shared difficulty is captured by a single parameter, v , the probability that a bad outcome is verified, whether inspection is carried out by the principal himself or, indirectly, by a court asked to enforce a bond. It is not obvious whether liability’s ability to fix the signaling problem grows or shrinks as this shared verifiability degrades, or how the answer depends on the value of the task and on how much capital a genuine expert can put behind the signal.

This paper builds a small model to analyze these questions. Section 2 sets out a two-stage model. A principal first chooses whether to perform a task himself with an AI or to enter a market for delegation. If he delegates, he faces an intermediary of privately known type, screened through a liability bond whose enforcement depends on the same verification technology that governs the principal’s own inspection. Section 3 solves the baseline model, setting aside whether a genuine expert can afford the bond the market requires. Its two building blocks, a make-or-buy threshold

in verification (Section 3.1) and a least-cost separating bond whose value to the principal does not depend on the enforcement probability (Section 3.2), are simple Spence-style results on their own. Nesting them (Section 3.3) shows when a signal is needed and when the whole apparatus is moot. Section 4 adds a capital constraint on bonding and delivers the main result. Write μ for the prior probability that the intermediary is the genuine type. The region of (v, μ) space in which genuine expertise cannot separate itself is non-empty exactly when the price exceeds a closed-form threshold. A second, higher threshold governs whether there are parameters at which the genuine expert cannot be engaged at all. Neither threshold depends on the quality of the genuine type. Section 5 discusses what this implies for responses to AI-driven signal erosion, Section 6 describes extensions, and Section 7 concludes.

Related literature. The analytical core combines a Spence (1973) and Riley (1979) signaling game with the Cho and Kreps (1987) intuitive criterion, applied to a liability bond rather than to education. The wealth constraint on bonding in Section 4 is the judgment-proof problem (Shavell, 1986) applied to the credibility of a signal rather than to ex post compensation. Two recent papers share this paper’s concern with AI, verification, and liability but locate the friction at a different stage. Gans (2026) studies staged access and liability for dual-use AI as a regulatory design problem facing a defender and an adversary, where the friction is release timing rather than delegation. Bryan and Gans (2026) study how an AI model should be trained given how humans will use it, a friction upstream of the market-structure question asked here. The friction studied here is whether a liability signal can survive once the technology that created the need for it also erodes a court’s ability to enforce it. It is closest in spirit to the credence-goods literature following Akerlof (1970) and Darby and Karni (1973), applied to a setting in which AI makes quality both cheap to fake and hard to verify.

2 Model

2.1 Setup and timing

A risk-neutral principal faces a task worth S if delivered good and $S - L$ if delivered bad, where $L > 0$ is the stake destroyed by a bad outcome. Two objects are drawn before either party acts:

- the task’s *verification level* $v \in [0, 1]$, the probability that inspection, if performed, reveals the true state. This is the same primitive whether inspection is carried out by the principal (DIY) or, indirectly, by a court asked to enforce a bond (delegation);
- the intermediary’s *type* $\theta \in \{H, L\}$, where H is a genuine expert and L is a *wrapper* reselling standard AI access. The type is private to the intermediary; the common prior is $\mu = \Pr(H)$.

Timing.

0. Nature draws v (common knowledge, Assumption 2) and θ (private to the intermediary, prior μ).

1. The principal chooses **DIY**, operating the AI himself and obtaining a good outcome with probability p_0 , or enters the **delegation market**.
2. If he delegates, the matched intermediary posts a bond $B(\theta) \geq 0$, subject to the capacity bound of Assumption 4. The principal observes B but not θ , forms a belief $\mu(B)$, and chooses to **engage** the intermediary at the exogenous price t or to **walk away**, falling back to DIY.
3. If engaged, the outcome is good with probability $p_\theta \equiv 1 - f_\theta$. If it is bad (probability f_θ), it is verified with probability $e(v)$, and B is forfeited to the principal; otherwise the loss goes uncompensated.

DIY's payoff, which incorporates the verify-and-retry mechanism of Assumption 3 below, is

$$U_{\text{DIY}}(v) = S - (1 - v)f_0L - c_0 - vf_0\kappa, \quad (1)$$

where $f_0 \equiv 1 - p_0$ is the principal's own (non-expert) failure probability, c_0 his cost of operating the AI directly, and κ the cost of a successful retry when inspection catches a bad draw. If the principal engages an intermediary of type θ that has posted bond B , expected payoffs are

$$\pi(\theta, B) = t - f_\theta e(v) B \quad (\text{intermediary}), \quad U(\theta, B) = S - t - f_\theta L + f_\theta e(v) B \quad (\text{principal}). \quad (2)$$

Write $\bar{U}$ for the principal's outside option. Section 3.2 treats $\bar{U}$ as a fixed number; from Section 3.3 on, $\bar{U} = U_{\text{DIY}}(v)$. Types are ordered by $f_L > f_H > 0$, and throughout we maintain $f_L L > t$: a wrapper's expected loss exceeds the fee. The principal engages when indifferent.

2.2 Assumptions

Assumption 1 (Fixed price). *The price t is exogenous and common to both intermediary types.*

Type affects the probability of success, not the price of operating the AI, consistent with the premise that raw AI execution is itself commoditized.

Assumption 2 (Common knowledge of verifiability). *The verification level v is known to both parties before the bond is posted.*

Assumption 3 (Mode-specific remedy). *A caught bad draw is remedied differently depending on who operates the AI. Under DIY, the principal redoes the work at cost κ . Under delegation, the principal cannot redo the intermediary's work; his remedy for a caught bad draw is the bond.*

Retry is the principal's self-correction technology; the bond is delegation's remedy. The distinction reflects what each party brings: the principal can redo his own work when he catches a failure, while what an intermediary offers is a higher probability of getting it right the first time.

Assumption 4 (Capacity bound). *The genuine type cannot credibly post a bond larger than $\bar{w}$.*

This is the judgment-proof problem applied to the credibility of the signal itself, not merely to ex post compensation. Only the genuine type’s capacity is bounded; we adopt this asymmetry for simplicity, since the results below concern the genuine expert’s ability to post a bond. Assumption 4 is set aside in Section 3 and imposed in Section 4.

Assumption 5 (Equilibrium selection). *Among perfect Bayesian equilibria that survive the intuitive criterion, we study (i) pooling without a bond when it satisfies the principal’s participation constraint; otherwise (ii) the least-cost separating equilibrium, when it exists; otherwise (iii) pooling at the smallest positive bond that sustains engagement, when one exists.*

Rule (ii) is a selection, not a refinement result. For priors close to the pooling threshold of Proposition 4, pooling at a positive bond can coexist with separation. Both intermediary types would prefer that outcome; the principal would not, since it holds him to his outside option.

The enforcement technology is taken in its simplest form, $e(v) = v$. This is the most direct reading of “if it is hard for the principal to verify, it is hard for a court to verify too” as one primitive rather than a correlated pair. A general affine specification, $e(v) = e_0 + (1 - e_0)v$, is examined in Appendix A.9 and does not affect the central result.

3 Baseline results

This section derives everything that follows without the capacity bound of Assumption 4. The baseline already shows when a signal is needed and when the whole apparatus is moot. On its own, however, it cannot generate a region in which genuine expertise is shut out, because nothing yet prevents a genuine expert from posting whatever bond separation requires. Section 4 changes that.

3.1 When does the principal do it himself?

Before introducing type uncertainty and a liability bond, it is useful to isolate the paper’s first intuition. Suppose for now that the intermediary is of a single, known type that succeeds with probability p_1 , to be compared with the principal’s own success probability p_0 . For a given verification level v , does the principal prefer to delegate to this known type at the posted price, or to do the task himself?

Proposition 1. *Suppose there is no uncertainty about the delegate’s type (its success probability p_1 is known) and $L > \kappa$. There is a unique threshold*

$$v^* = \frac{\Delta p \cdot L - \Delta c}{f_0(L - \kappa)}, \quad \Delta p \equiv p_1 - p_0, \quad \Delta c \equiv t - c_0, \quad (3)$$

below which delegating at the posted price makes the principal strictly better off than DIY, and above which delegation cannot make him better off than DIY.

Proof. Appendix A.1. □

High verifiability makes catch-and-retry a substitute for expertise: if a bad draw can be cheaply caught and redone, the principal does not need someone who gets it right the first time. Low verifiability makes the retry option worthless, because the principal cannot tell good from bad. The comparison then collapses to whether the quality premium is worth the price premium, which is exactly the terrain on which a credence-good unraveling problem can bite. The threshold is not yet about liability. It establishes that verification technology, not production capability, is what should predict which tasks migrate to direct AI use.

3.2 The liability bond as a signal

Now introduce type uncertainty and set the DIY question aside for the moment. The principal faces an intermediary of unknown type θ , screened through a bond, at a fixed enforcement probability e and a fixed outside option $\bar{U}$.

Proposition 2. *Suppose a known wrapper is not worth engaging even at a zero fee, $S - f_L L \leq \bar{U}$. Then the least-cost bond that separates H from L is*

$$B_H^* = \frac{t}{f_L e}, \quad (4)$$

and H is always willing to post it. It sustains engagement whenever the principal prefers an identified H at this bond to his outside option.

Proof sketch. The marginal cost of posting bond is $f_\theta e$, strictly lower for H since $f_L > f_H$. Single crossing is therefore derived from one type-specific primitive, each type's own probability of a bad draw (f_θ), rather than assumed. A wrapper that reveals itself can obtain engagement only by posting a bond that leaves the principal at least $\bar{U}$, which leaves the wrapper at most $S - f_L L - \bar{U} \leq 0$; its payoff when revealed is therefore zero. Mimicking H's bond is thus deterred exactly when $t - f_L e B \leq 0$, and the least such bond is B_H^* . H's own payoff, $t - f_H e B_H^* = t(1 - f_H/f_L)$, is positive. Full derivation in Appendix A.2. $\square$

Proposition 3. *At B_H^* , the principal's payoff does not depend on e : $\partial U(H, B_H^*)/\partial e = 0$, provided the enforcement technology is costless to run. Any real, additive enforcement or bond-carrying cost breaks this invariance.*

Proof. One substitution: $U(H, B_H^*) = S - t - f_H L + f_H e \cdot t/(f_L e) = S - t - f_H L + f_H t/f_L$, in which e cancels between the grossed-up bond and the recovery term. $\square$

The result is easy to over-read, so it is worth stating precisely. Worse enforcement inflates the *nominal* bond needed to keep deterring the wrapper ($\partial B_H^*/\partial e < 0$), and that inflation exactly offsets the lower probability of collecting it. Expected recovery, and with it the principal's payoff, is untouched. Better courts, in this model, do not make separation more valuable once it is achieved; they make it more *feasible*, provided it is achievable at all — a conclusion that assumes costless bonding, revisited under more realistic costly bonds in Section 6. That qualifier, whether the bond

can actually be posted, is what the capacity bound of Assumption 4 puts back into the model, and it is where Section 4 picks up.

Proposition 4. *Pooling at $B = 0$ satisfies the principal's participation constraint if and only if $\mu \geq \mu^*$, where*

$$\mu^* = \frac{f_L - \frac{S-t-\bar{U}}{L}}{f_L - f_H}. \quad (5)$$

Proof. Appendix A.3. □

Is a signal needed at all, or does the pooled belief already satisfy the principal's participation constraint without costly bonding? This is a genuine gate, separate from whether a separating bond, once needed, is affordable. Above μ^* , H has no reason to signal: engaged at $B = 0$, it already earns t , and any bond only lowers that to $t - f_H eB$. This follows from Assumption 1. Because the price is fixed and common to both types, being identified as H is worth nothing beyond being engaged, unlike in Spence (1973), where the wage itself responds to beliefs. With a belief-contingent price, H would want to separate even above μ^* . The threshold rises with the principal's outside option and with the price, and (whenever $\mu^* > 0$) falls as the genuine type improves, because a better genuine type makes the pooled market more attractive.

3.3 Nesting the two: the baseline map

Substitute $\bar{U} = U_{\text{DIY}}(v)$ from (1) and $e(v) = v$ into Propositions 2–4. This is the paper's basic interaction: a delegation market screened by a bond, evaluated against an outside option that itself depends on the same verification parameter. Write

$$\Delta \equiv Lf_0 + c_0 - Lf_L \quad (6)$$

for the principal's expected shortfall, at $v = 0$, relative to a known wrapper working for free. The scope condition of Proposition 2, $S - f_L L \leq U_{\text{DIY}}(v)$, then reads $v \geq v_W \equiv \Delta/[f_0(L - \kappa)]$.

Proposition 5. *Suppose $L > \kappa$. The pooling threshold and the participation boundary for an identified genuine expert are*

$$\mu^*(v) = \frac{f_L - f_0 + \frac{t-c_0}{L} + v \frac{f_0(L-\kappa)}{L}}{f_L - f_H}, \quad v_{\text{IR}} = \frac{f_0 L + c_0 - t - f_H L + f_H t/f_L}{f_0(L - \kappa)}, \quad (7)$$

with $\mu^*(v)$ affine and increasing in v . The (v, μ) plane divides into three regions:

1. $\mu \geq \mu^*(v)$: pooling, no bond needed.
2. $\mu < \mu^*(v)$, $v \leq v_{\text{IR}}$: a positive bond is needed for engagement. For $v \geq v_W$, H separates at B_H^* ; for $v < v_W$, the two types pool at a positive bond.
3. $v > v_{\text{IR}}$: delegation cannot make the principal better off than DIY.

Proof. Appendix A.4. □

Region 3 is Proposition 1's threshold reappearing once type uncertainty and a bond are introduced, and the correspondence is exact:

$$v_{\text{IR}} = v^*|_{f_1=f_H} + \frac{f_H t / f_L}{f_0(L - \kappa)}. \quad (8)$$

That is, v_{IR} is the make-or-buy threshold evaluated at the genuine type, shifted up by the expected bond recovery $f_H e B_H^* = f_H t / f_L$ (the e -invariant object of Proposition 3) divided by the slope of U_{DIY} in v . In short, v_{IR} is Proposition 1 plus the bond. At high verifiability the signaling apparatus is moot. Region 1 says that a signal is not always needed even at moderate verifiability, if the prior is favorable enough. Region 2 is where the market for genuine expertise needs a bond and operates. Its lower-verifiability part, $v < v_W$, is where a known wrapper is itself worth engaging at a high enough bond, so the wrapper cannot be screened out and the two types trade at a common bond. Nothing in this baseline map says that the genuine expert can be shut out. That is because, without Assumption 4, H can post whatever bond is required. Section 4 removes that fiction.

4 Capital-constrained bonding

We now impose Assumption 4. It refines region 2 of Proposition 5 into two parts: the bond genuine expertise would need to post is either affordable or it is not, and only once the second possibility is on the table can the genuine expert be shut out.

Proposition 6. *Separation at B_H^* is feasible only if $B_H^* \leq \bar{w}$, that is, $v \geq v_{\text{cap}} \equiv t / (f_L \bar{w})$. Under $e(v) = v$, the map of Proposition 5 refines to four regions:*

1. $\mu \geq \mu^*(v)$: pooling, no bond needed.
2. $\mu < \mu^*(v)$, $v_{\text{cap}} \leq v \leq v_{\text{IR}}$: the required bond is affordable; outcomes as in region 2 of Proposition 5.
3. $\mu < \mu^*(v)$, $v < v_{\text{cap}}$, $v \leq v_{\text{IR}}$: separation at B_H^* is infeasible on capacity grounds.
4. $v > v_{\text{IR}}$: delegation cannot make the principal better off than DIY.

Proof. Appendix A.5. □

As v falls, the bond needed to separate rises (Proposition 2) while the DIY fallback deteriorates (Proposition 1). These two forces push the width of region 2 in opposite directions, and their net effect cannot be signed by inspection. Region 3 is new relative to the baseline map, and it is exactly where genuine expertise would be most valuable to identify and cannot credibly signal itself. The rest of this section asks whether region 3 is ever non-empty and, if so, whether the genuine expert is actually shut out there.

4.1 When does the market fail?

Proposition 7. *Suppose $L > \kappa$, and let v_0 solve $\mu^*(v_0) = 0$. Provided $v_0 < 1$, region 3 is non-empty (there is $v \in (0, 1)$ with $\mu^*(v) > 0$ and $v < v_{\text{cap}}$) if and only if*

$$t > \hat{t} \equiv \frac{f_L \bar{w} \Delta}{f_0(L - \kappa) + f_L \bar{w}}. \quad (9)$$

Proof. Appendix A.6. □

Proposition 8. *Region 3 is not, on its own, a region in which the genuine expert is shut out: pooling at a bond within the expert's capacity remains feasible over part of it. Suppose $L > \kappa$, $\Delta > t$ and $v_{\text{cap}} < \min\{v_{\text{IR}}, 1\}$. There are parameters in region 3 at which the genuine expert cannot be engaged in any separating or pooling equilibrium if and only if*

$$t > \hat{t}_{\text{pool}} \equiv \frac{f_L \bar{w} \Delta}{f_0(L - \kappa)} > \hat{t}. \quad (10)$$

Proof. Appendix A.7. □

The correction from Proposition 7 to Proposition 8 is substantive. In the numerical example of Appendix A.8, at $t = 1.5$ and $v = 0.5$, pooling at an affordable bond rescues 53% of region 3's height. Figure 1 shows the full map for the same example at $t = 1.2$.

Proposition 9 (Central result). *The thresholds $\hat{t}$ and $\hat{t}_{\text{pool}}$ do not depend on f_H :*

$$\frac{\partial \hat{t}}{\partial f_H} = 0, \quad \frac{\partial \hat{t}_{\text{pool}}}{\partial f_H} = 0. \quad (11)$$

Proof. $\hat{t}$ is the price at which $v_0 = v_{\text{cap}}$, and $\hat{t}_{\text{pool}}$ is the price at which $v_W = v_{\text{cap}}$ (Appendices A.6 and A.7). Neither $v_{\text{cap}} = t/(f_L \bar{w})$ nor $v_W = \Delta/[f_0(L - \kappa)]$ contains f_H . v_0 is the root of $\mu^*(v)$, and f_H enters $\mu^*(v)$ only through the positive factor $1/(f_L - f_H)$, which stretches the function but does not move its root. Robustness to more general enforcement technologies is in Appendix A.9. □

This is the result the paper is organized around. **Whether the market for genuine expertise can fail at all does not depend on how good genuine expertise is.** It depends on the quality of the worst type in the market (f_L), the principal's own fallback (f_0, c_0, κ), and how much capital a genuine expert can post ($\bar{w}$), besides the stake L , and never on f_H . Improving the best available experts, through better training or better tools, does nothing to prevent the failure region from existing.

The direction of the remaining margins is worth stating, because one of them runs against intuition. More bonding capacity always helps: with $\Delta > 0$, both thresholds increase in $\bar{w}$. A better fallback for the principal works the other way. Lowering c_0 or κ lowers both thresholds, and so does lowering f_0 whenever $L f_L > c_0$. A principal who does better on his own demands more from any delegated arrangement, so the pooled market fails to satisfy his participation constraint more

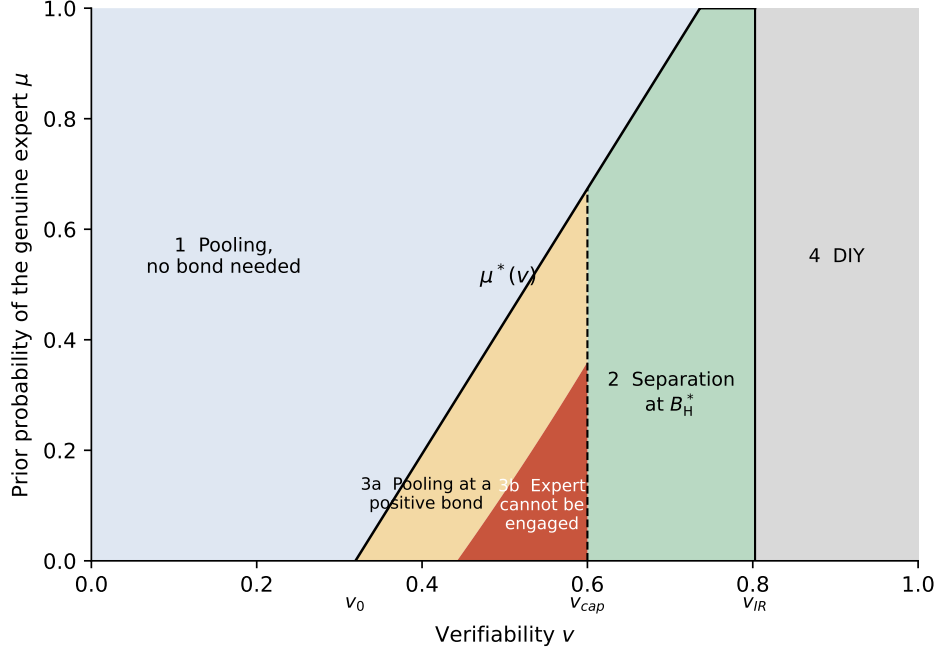


Figure 1: The (v, μ) map of Proposition 6 for the example of Appendix A.8 at $t = 1.2$, where $\hat{t} = 0.761$ and $\hat{t}_{\text{pool}} = 0.972$. Region 1 lies above $\mu^*(v)$; region 4 lies to the right of $v_{\text{IR}} = 0.803$. Region 3 lies below $\mu^*(v)$ and to the left of $v_{\text{cap}} = 0.600$, where the least-cost separating bond exceeds the genuine expert’s capacity. In its upper part (3a), pooling at an affordable bond still sustains engagement. In its lower part (3b), the genuine expert cannot be engaged at any bond it can afford.

often, and the failure region appears at lower prices. The quality of the worst type enters through opposing channels and has no general sign.

Scope matters as well: this is a result about *existence*, not extent. The height of the failure region scales with $1/(f_L - f_H)$, so as $f_H \rightarrow f_L$ (as the genuine type becomes barely distinguishable from a wrapper) region 3 comes to cover every prior for v between v_0 and v_{cap} . f_H does not decide whether the market can fail; it decides how much of the space it fails over, once it can.

5 Discussion

Four intuitions motivated this paper and were, at the outset, in tension without a resolution mechanism. Genuine expertise commands a premium. That premium is hard to verify once AI execution is a commodity. Liability can restore the signal, since a bond substitutes for direct verification. But enforcement of that liability is hostage to the same verification problem the bond exists to route around. The model shows *when* these forces cancel rather than merely asserting that they are in tension. The baseline of Section 3 shows that verifiability, once separation is achieved, matters only through feasibility and not through the size of the payoff (Proposition 3). On its own it cannot say when feasibility fails, because nothing in it stops a genuine expert from posting whatever bond deterrence requires. Section 4 supplies the missing piece by imposing the capital constraint of

Assumption 4. At low verifiability, the DIY fallback deteriorates while the bond needed to sustain separation inflates against a *fixed* capacity ceiling, and Proposition 7 shows that these need not offset. There is a closed-form region of (v, μ) space in which genuine expertise cannot credibly signal itself, and it sits where that expertise would be most valuable to identify. In substance this is the judgment-proof problem (Shavell, 1986), applied to the credibility of a signal rather than to ex post compensation, which is why it interacts with the verification parameter v rather than standing alone.

The policy-relevant reading of Proposition 9 differs from an intuition already available informally. A regulator or professional body worried about AI eroding the market for expert judgment might first reach for interventions that raise the ceiling, such as higher training standards or better tools for the most capable practitioners. Proposition 9 says this is not where the failure region’s existence is decided. Existence is decided by the bottom of the market, by the principal’s own fallback, and by the capital an expert can put behind a bond. Of these, only bonding capacity has an unambiguous direction, which points to interventions that expand the capital a genuine expert can credibly commit. The fallback result carries a specific warning for the AI context. As AI tools improve the principal’s own attempt, the delegation market must clear a higher bar, and the region in which genuine expertise cannot credibly signal itself appears at lower prices.

One assumption deserves a direct flag rather than a footnote. Assumption 5 selects the separating equilibrium wherever it exists, even where a Pareto-superior pooling equilibrium also survives the intuitive criterion. That selection has a substantive justification beyond the refinement gap: pooling clears the principal’s participation constraint only in expectation, and a principal who cannot restore the underlying harm by a monetary transfer, or who is evaluated ex post rather than ex ante, has reason to reject it even when $\bar{U}$ is met on average. Section 6 returns to this.

6 Extensions

- **Costly bonds.** Real bonds carry costs roughly proportional to their face value, such as surety premiums, letter-of-credit fees, or the opportunity cost of collateral. Since the face value of B_H^* scales with $1/e$, weaker enforcement then makes separation more wasteful even when it is affordable, and the invariance of Proposition 3 no longer holds.
- **Endogenous verification.** Verifiability v is a task-level primitive here. Letting the principal invest in his own verification capacity opens a second under-investment problem, in the capacity to *check* alongside the capacity to *produce*.
- **Correlated failure and bond credibility.** The capacity ceiling $\bar{w}$ is exogenous here. A large firm that posts a large nominal bond it could not honor in a correlated failure is a distinct threat to the signal’s credibility.
- **Liability as one signal among several.** The analysis concerns a liability bond. Reputation, credentials, and institutional affiliation are alternative signals with verification requirements of

their own, and AI erodes them through different channels. The paper does not compare them.

- **Risk and the ex-post cost of pooling.** The expected-value participation constraint suits the intermediary, whose bond-posting decision is naturally evaluated over repeated trials, but is a strong assumption for a one-shot principal, particularly where harm is not restorable by a transfer, as with a misdiagnosis rather than a mispriced audit. It also obscures an ex-post fact that expectation cannot see: pooling does not reveal to any given principal which type he engaged, so some fraction of the pool bears the full harm from a wrapper the bond never separated out. Replacing the expected-value participation constraint with a risk-averse or ex-post welfare criterion is left for future work.

7 Conclusion

Two mechanisms jointly determine whether a market for genuine expertise survives the arrival of cheap, competent AI execution: the principal’s ability to catch and redo a bad result himself, and a bond that lets genuine expertise be credibly signaled when he cannot. This paper shows that whether that market can fail is not decided by how good the best available expert is. It is decided by the quality of the worst imitator in the market, the principal’s own unaided fallback, and how much capital a legitimate expert can put behind a bond. Bonding capacity is the margin that works unambiguously in the market’s favor. A better fallback, which is exactly what AI gives the principal, makes the failure region appear at lower prices. The model studies liability as the signal, takes verification as a task-level given, and fixes the price. Within those limits the result is exact: making the best experts better, with nothing else changing, does not prevent the market for genuine expertise from failing.

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A Proofs

A.1 Proof of Proposition 1

With the intermediary’s type known there is nothing to signal. At the posted price and with no bond, the delegated payoff is (2) at the known type with $B = 0$, and, by Assumption 3, it has no retry term:

$$U_{\text{DEL}} = S - t - f_1 L, \quad f_1 \equiv 1 - p_1,$$

which does not depend on v . The DIY payoff (1) is affine in v :

$$U_{\text{DIY}}(v) = S - f_0 L - c_0 + v f_0 (L - \kappa).$$

Their difference is

$$U_{\text{DEL}} - U_{\text{DIY}}(v) = \Delta p \cdot L - \Delta c - v f_0 (L - \kappa), \quad \Delta p = f_0 - f_1, \quad \Delta c = t - c_0.$$

Given $L > \kappa$ the slope is strictly negative, so the difference changes sign exactly once, at v^* in (3). Below v^* , delegating at the posted price is strictly better than DIY. Above v^* , the principal walks away at $B = 0$. A known intermediary could then post a bond, which acts as an expected transfer to the principal, but the smallest bond that secures engagement leaves the principal exactly at $U_{\text{DIY}}(v)$. Delegation therefore cannot make him better off than DIY. (If $\kappa > L$ the slope reverses and so does the direction of the comparison; the case $L > \kappa$ is maintained throughout.)

A.2 Proof of Proposition 2

The marginal cost to type θ of posting bond B is $\partial(f_\theta e B)/\partial B = f_\theta e$, independent of B . Costs are linear, so single crossing holds trivially: H’s cost lies strictly below L’s at every $B > 0$.

Consider a wrapper that is identified as such. The principal engages it at bond B if and only if $S - t - f_L L + f_L e B \geq \bar{U}$. At the smallest such bond, the wrapper earns

$$t - f_L e B = t - (\bar{U} - S + t + f_L L) = S - f_L L - \bar{U},$$

and any larger bond earns it less. Under $S - f_L L \leq \bar{U}$ an identified wrapper therefore cannot earn a positive payoff, and its payoff when separated is 0. Mimicking a bond B that identifies H is deterred if and only if

$$t - f_L e B \leq 0 \iff B \geq \frac{t}{f_L e}.$$

H wants the smallest bond satisfying this, since bonding is costly, so $B_H^* = t/(f_L e)$. Its payoff there is $t - f_H e B_H^* = t(1 - f_H/f_L) > 0$. Engagement at B_H^* additionally requires $U(H, B_H^*) \geq \bar{U}$. Separating equilibria at bonds above B_H^* fail the intuitive criterion: a deviation to a bond between B_H^* and the equilibrium bond is equilibrium-dominated for L but not for H, so the principal's belief must place it on H, and H gains by deviating. Off-path bonds below B_H^* can be attributed to L, since L would gain from them if engaged. Hence, as in Riley (1979), B_H^* is the unique separating bond that survives the intuitive criterion.

A.3 Proof of Proposition 4

When both types pool at $B = 0$, the principal's posterior equals the prior, and his payoff is

$$U_{\text{pool}}(\mu) = S - t - [\mu f_H + (1 - \mu) f_L] L,$$

which is increasing in μ since $f_H < f_L$. Setting $U_{\text{pool}}(\mu) = \bar{U}$ and solving for μ gives (5), and $\mu \geq \mu^*$ satisfies the constraint. The comparative statics in the text follow from $\partial \mu^* / \partial \bar{U} = \partial \mu^* / \partial t = 1/[L(f_L - f_H)] > 0$ and $\partial \mu^* / \partial f_H = \mu^* / (f_L - f_H)$.

A.4 Proof of Proposition 5

Pooling threshold. Substituting $\bar{U} = U_{\text{DIY}}(v)$ into (5),

$$S - t - U_{\text{DIY}}(v) = (c_0 - t) + (1 - v) f_0 L + v f_0 \kappa,$$

so

$$\mu^*(v) = \frac{f_L - \frac{(c_0 - t) + (1 - v) f_0 L + v f_0 \kappa}{L}}{f_L - f_H} = \frac{f_L - f_0 + \frac{t - c_0}{L} + v \frac{f_0(L - \kappa)}{L}}{f_L - f_H} = \frac{v f_0(L - \kappa) - \Delta + t}{L(f_L - f_H)}.$$

This is affine in v with slope $f_0(L - \kappa)/[L(f_L - f_H)] > 0$ when $L > \kappa$.

Participation boundary. By Proposition 3, $U(H, B_H^*) = S - t - f_H L + f_H t / f_L$ does not depend on v , while $U_{\text{DIY}}(v)$ is increasing. They cross once, at v_{IR} in (7); the principal prefers an identified H at B_H^* to DIY if and only if $v \leq v_{\text{IR}}$. Identity (8) follows by evaluating (3) at $f_1 = f_H$ and comparing numerators.

Ordering. Let v_0 solve $\mu^*(v_0) = 0$, so $v_0 = (\Delta - t)/[f_0(L - \kappa)]$. Then

$$v_W - v_0 = \frac{t}{f_0(L - \kappa)} > 0, \quad v_{\text{IR}} - v_W = \frac{(f_L - f_H)(L - t/f_L)}{f_0(L - \kappa)} > 0,$$

the second since $f_L L > t$. Hence $v_0 < v_W < v_{\text{IR}}$. Moreover, $\mu^*(v_{\text{IR}}) = 1 + (t/L)[1/(f_L - f_H) - 1/f_L] > 1$, so region 1 does not intersect $v \geq v_{\text{IR}}$.

Region 1. For $\mu \geq \mu^*(v)$, pooling at $B = 0$ satisfies the principal's participation constraint (Proposition 4). No type gains by deviating to a positive bond, since it is already engaged at t . This

is the outcome selected by Assumption 5(i).

Region 2. For $\mu < \mu^*(v)$, pooling at $B = 0$ fails, so engagement requires a positive bond. If $v_W \leq v \leq v_{IR}$, the scope condition of Proposition 2 holds and $U(H, B_H^*) \geq U_{DIY}(v)$, so H separates at B_H^* . If $v < v_W$, an identified wrapper can obtain engagement and earn $S - f_L L - U_{DIY}(v) > 0$, so no separating equilibrium in which the wrapper earns zero exists. Consider instead pooling at the bond that leaves the principal indifferent at belief μ ,

$$B_p^*(\mu; v) = \frac{U_{DIY}(v) - S + t + \bar{f}L}{\bar{f}v}, \quad \bar{f} \equiv \mu f_H + (1 - \mu)f_L.$$

For $v \leq v_{IR}$ we have $S - t - U_{DIY}(v) \geq f_H(L - t/f_L) > 0$, so B_p^* is decreasing in μ , and $B_p^*(0; v)$ is the bond at which the principal is willing to engage an identified wrapper. The wrapper's payoff in the pool, $t - f_L v B_p^*(\mu; v)$, is therefore at least its payoff from being identified, $t - f_L v B_p^*(0; v)$. Since $B_p^*(0; v) - B_H^* = [v f_0(L - \kappa) - \Delta]/(f_L v) < 0$ for $v < v_W$, the pooled bond is below B_H^* and both types earn a positive payoff. Deviations to smaller bonds may be attributed to L, at which the principal walks away; deviations to larger bonds are unprofitable for both types. The pool is an equilibrium, and it is the outcome of Assumption 5(iii).

Region 3. Let $v > v_{IR}$. Any bond $B \leq B_H^*$ that both types might post yields the principal $S - t - \bar{f}(L - eB)$, which is at most $S - t - f_H(L - eB_H^*) = U(H, B_H^*) < U_{DIY}(v)$, since $eB \leq t/f_L < L$. A wrapper will not post a bond above B_H^* . An identified H can still obtain engagement by posting a bond above B_H^* , which acts as an expected transfer to the principal, but the smallest such bond leaves him exactly at $U_{DIY}(v)$. Delegation therefore cannot make the principal better off than DIY.

A.5 Proof of Proposition 6

Separation at B_H^* requires $B_H^* \leq \bar{w}$. With $e(v) = v$, $B_H^* = t/(f_L v) \leq \bar{w}$ if and only if $v \geq v_{cap} = t/(f_L \bar{w})$, a vertical line in (v, μ) space. Region 2 of Proposition 5 therefore splits at v_{cap} . For $v_{cap} \leq v \leq v_{IR}$, the separating bond is affordable; so is the pooled bond when $v < v_W$, since it lies below B_H^* . For $v < v_{cap}$, separation at B_H^* is infeasible whether or not the principal's participation constraint would otherwise hold. Regions 1 and 4 are unchanged, since neither $\mu^*(v)$ nor v_{IR} involves $\bar{w}$. Since $v_{IR} - v_0 = [L(f_L - f_H) + f_H t/f_L]/[f_0(L - \kappa)] > 0$, points just above v_0 always satisfy $v \leq v_{IR}$.

A.6 Proof of Proposition 7

Given $L > \kappa$, $\mu^*(v)$ is increasing, so $\mu^*(v) > 0$ if and only if $v > v_0$. Region 3 requires $\mu^*(v) > 0$ and $v < v_{cap}$ simultaneously, that is, a non-empty interval (v_0, v_{cap}) , or $v_0 < v_{cap}$. Clearing positive denominators,

$$\frac{t}{f_L \bar{w}} > \frac{\Delta - t}{f_0(L - \kappa)} \iff t[f_0(L - \kappa) + f_L \bar{w}] > f_L \bar{w} \Delta \iff t > \hat{t}.$$

Since $v_0 < v_{\text{IR}}$, points just above v_0 also satisfy $v \leq v_{\text{IR}}$. The proviso $v_0 < 1$ ensures that the interval intersects $[0, 1]$; the inequality $v_0 < v_{\text{cap}}$ alone does not guarantee this.

A.7 Proof of Proposition 8

In region 3, $B_{\text{H}}^* > \bar{w}$. The genuine expert can be engaged in a pooling equilibrium at (v, μ) if and only if $B_p^*(\mu; v) \leq \bar{w}$; the wrapper then joins the pool, by the argument of Appendix A.4, since $B_p^*(\mu; v) \leq \bar{w} < B_{\text{H}}^*$ and $B_p^*(\mu; v) \leq B_p^*(0; v)$. The expert can be separated from the wrapper only at a bond of at least B_{H}^* when $v \geq v_{\text{W}}$, or of at least $B_p^*(0; v)$ when $v < v_{\text{W}}$ (the bond at which an identified wrapper is engaged). Since B_p^* is decreasing in μ , the genuine expert cannot be engaged at some prior μ close to 0 at verification level v if and only if

$$B_p^*(0; v) = \frac{f_0(L - \kappa)v - (\Delta - t)}{f_{\text{L}}v} > \bar{w}.$$

When $\Delta > t$, $B_p^*(0; v)$ is increasing in v , so over $v < v_{\text{cap}}$ this holds somewhere if and only if it holds in the limit $v \rightarrow v_{\text{cap}}$. Using $B_{\text{H}}^*(v_{\text{cap}}) = \bar{w}$ and $B_p^*(0; v) - B_{\text{H}}^*(v) = [f_0(L - \kappa)v - \Delta]/(f_{\text{L}}v)$, the condition is $f_0(L - \kappa)v_{\text{cap}} > \Delta$, i.e. $v_{\text{cap}} > v_{\text{W}}$, which is equivalent to

$$t > \hat{t}_{\text{pool}} = \frac{f_{\text{L}}\bar{w}\Delta}{f_0(L - \kappa)}.$$

Points just below v_{cap} lie in region 3: they satisfy $v \leq v_{\text{IR}}$ by assumption, and $\mu^*(v) > 0$ there because $t > \hat{t}_{\text{pool}} > \hat{t}$ implies $v_0 < v_{\text{cap}}$. Finally, $\hat{t}_{\text{pool}} > \hat{t}$ because the two share a numerator and $\hat{t}$'s denominator is larger by $f_{\text{L}}\bar{w} > 0$, with $\Delta > t > 0$.

A.8 Numerical example

Let $f_{\text{H}} = 0.2$, $f_{\text{L}} = 0.5$, $f_0 = 0.8$, $L = 10$, $\kappa = 1$, $c_0 = 0.5$, $\bar{w} = 4$, and $S = 20$. Then $\Delta = 3.5$, $\hat{t} = 0.761$, and $\hat{t}_{\text{pool}} = 0.972$.

Table 1: Region boundaries in the numerical example

t	v_0	v_{cap}	v_{IR}	v_{W}	Region 3 non-empty	Expert shut out somewhere
0.50	0.417	0.250	0.861	0.486	no	no
0.80	0.375	0.400	0.836	0.486	yes	no
1.20	0.319	0.600	0.803	0.486	yes	yes
1.50	0.278	0.750	0.778	0.486	yes	yes

At $t = 0.8$, region 3 is non-empty, yet pooling at an affordable bond sustains engagement throughout it, which illustrates why Proposition 8 matters. At $t = 1.5$, $v_0 < v_{\text{cap}} < v_{\text{IR}} < 1$, so all four regions of Proposition 6 are non-empty and interior. At $t = 1.5$ and $v = 0.5$, $\mu^*(0.5) = 0.533$, and pooling at an affordable bond is feasible for all $\mu \geq 0.250$, that is, over 53% of region 3's height at that v . Figure 1 plots the case $t = 1.2$.

A.9 Robustness of Proposition 9

General affine enforcement, $e(v) = e_0 + (1 - e_0)v$. The capacity threshold becomes $v_{\text{cap}} = (\hat{e} - e_0)/(1 - e_0)$ with $\hat{e} = t/(f_L \bar{w})$, and the existence threshold of Proposition 7 generalizes to

$$\hat{t}_{e_0} = \frac{f_L \bar{w} [(1 - e_0)\Delta + e_0 f_0 (L - \kappa)]}{f_0 (L - \kappa) + f_L \bar{w} (1 - e_0)},$$

which reduces to $\hat{t}$ at $e_0 = 0$ and does not depend on f_H .

Dropping $L > \kappa$. If $\kappa > L$, $\mu^*(v)$ is decreasing, region 3 becomes $\{v < \min(v_0, v_{\text{cap}})\}$, and the existence condition simplifies to $t > \Delta$. This is a different, simpler threshold, and it is also independent of f_H .

In each case the mechanism is the same: f_H enters $\mu^*(v)$ only through the positive factor $1/(f_L - f_H)$, and none of v_{cap} , v_0 , or v_W contains f_H . The *size* of the failure region, by contrast, scales with $1/(f_L - f_H)$ and grows without bound as $f_H \rightarrow f_L$.